QUESTION 1

A space probe contains a series of fixed sensing arrays. These arrays are directed at different stellar objects during its journey so precise attitude (orientation) control is required. Rotation of the probe about each axis (x, y and z) is regulated by separate control systems of the same type. The block diagram for this type of control system is given in Fig. 1 below.

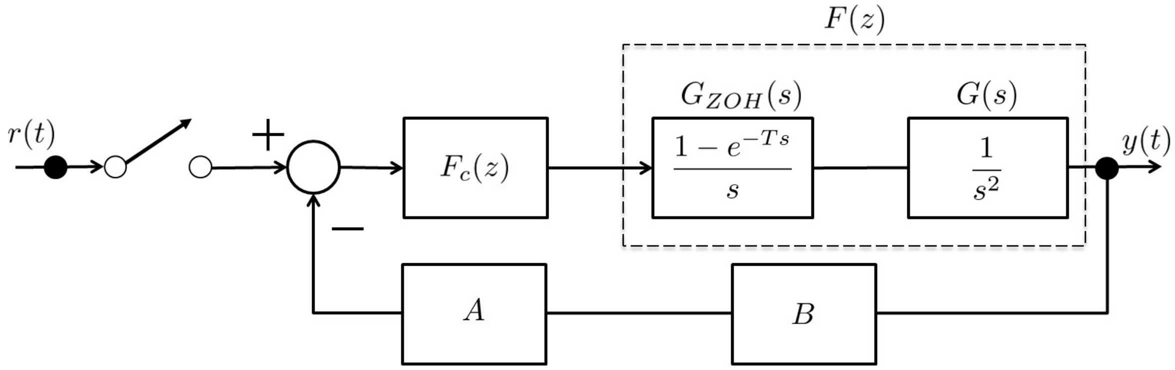

Figure 1: Probe Attitude Stabilisation System (single axis)

(a) Find the discrete time (pulse) transfer function $F(z)$ in Fig. 1 representing the station dynamics about a single axis assuming a sampling time $T = 0.2$ seconds.

(b) How does the sampling time T affect the stability properties of the system $F(z)$?

(c) Rotation about each axis can be regulated using a controller $G_c(s)$ given by (1) below

$$G_c(s) = K_c \frac{(s+1)}{(s+16)} \tag{1}$$

Find an equivalent discrete time controller $F_c(z)$ for $G_c(s)$ assuming gain K_c.

(d) From a practical perspective, suggest appropriate functions for the blocks A and B located in the feedback path of Fig. 1. How could you modify A or B to improve the control system performance?

QUESTION 2

A recycling company uses digital control hardware technology to help manage production of sustainable fuel products from organic waste. They have been using a set of difference equations given by (2) to represent the chemical reactions inside the tanks

$$\begin{aligned}\epsilon(kT+T) &= \epsilon(kT) - 2a^{-1}(1-e^{-aT})\theta(kT) && \text{(2a)}\\ \theta(kT+T) &= e^{-aT}\theta(kT) - a^{-1}(1-e^{-aT})u(kT) && \text{(2b)}\\ y(kT+T) &= -4a\theta(kT) && \text{(2c)}\end{aligned}$$

where θ is the temperature, ϵ is the fuel ethanol concentration, u is the organic waste input rate (kgs^{-1}) and y is the measured output. The company's fuel purity has degraded and believe their digital control systems are responsible.

(a) Provide a brief analysis of the model that would help the company design digital control systems to regulate ethanol concentration assuming a sampling time $T = 2$ seconds and parameter $a = 0.1$.

(b) The company asks you to design a digital control system to regulate ethanol concentration. They know the desired pole location for the equivalent continuous time system to (2) should be $p_{1,2} = 2/3 \pm 1/2j$. Design a state feedback controller to meet these specifications

(c) The company implements your controller, but the performance specifications are not achieved. You request more information and they provide the difference equations in (3).

$$\begin{aligned}\epsilon(kT+T) &= \epsilon(kT) - 3.30\theta(kT) && \text{(3a)}\\ \theta(kT+T) &= 0.67\theta(kT) - 1.65u(kT) && \text{(3b)}\end{aligned}$$

Diagnose the problem then suggest potential solutions for the company, justifying your response with evidence.

QUESTION 3

A sophisticated tool is used in knee surgery to bend around structures and make precise incisions. The tool can be modelled as a second order system with states α and γ related to the curvature of the end effector segment. The control input $u(t) = \delta_P(t)$ relates to fluid pressure within the device. The discrete time state space model for the system is given by (4)

$$\begin{bmatrix} \alpha(kT+T) \\ \gamma(kT+T) \end{bmatrix} = \begin{bmatrix} 0.6 & 0.4 \\ \text{-}0.9 & 0.6 \end{bmatrix} \begin{bmatrix} \alpha(kT) \\ \gamma(kT) \end{bmatrix} + \begin{bmatrix} 0 \\ \text{-}1.0 \end{bmatrix} \delta_P(kT) \tag{4}$$

(a) Design a digital state feedback controller for the system in (4) such that the closed loop system has a settling time $t_s = 1.9$ s and natural frequency $\omega_n = 3$ rads^{-1} assuming a sampling time $T = 0.5$ seconds.

(b) What assumptions have you made in part (a) and are they reasonable?

The company that manufactures the surgical tool wants to improve its performance. They propose two state-feedback controllers for the system in (4) denoted by $\delta_P^1(kT)$ and $\delta_P^1(kT)$ and given in (5) and (6) respectively.

$$\delta_P^2(kT) = 0.25\alpha(kT) - 0.99\beta(kT) \tag{5}$$
$$\delta_P^1(kT) = 0.19\alpha(kT) - 0.86\beta(kT) \tag{6}$$

(c) Which controller should the company use to improve the performance of the surgical tool? Justify your answer.

QUESTION 4

For maximum power generation, a simple one axis solar panel array must be orientated toward the sun. The array moves by generating a moment $u(t)$ about the arrays centre of mass. The angle of rotation $\theta(t)$ describes the relative orientation of the array to the sun. A simplified state-space model for the system dynamics is given by

$$\begin{bmatrix}\dot{\theta}(t)\\ \ddot{\theta}(t)\end{bmatrix} = \begin{bmatrix}0 & -1\\ -2 & -5\end{bmatrix}\begin{bmatrix}\theta(t)\\ \dot{\theta}(t)\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u(t) \tag{7a}$$

$$y(t) = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}\theta(t)\\ \dot{\theta}(t)\end{bmatrix} \tag{7b}$$

(a) Using a 2nd order approximation for the matrix exponential, and assuming sampling time $T = 1$ second, show that the discrete-time (pulse) transfer function $F(z)$ for the continuous system in (7) is given by (8) below.

$$F(z) = \frac{0.5z^{-1} - 1.5z^{-1}}{1 - 7.5z^{-1} + 4.5z^{-2}} \tag{8}$$

(b) Analyse the stability of the systems given by (7) and (8). What do you notice?

The solar panel array can be controlled using classical feedback control approaches and digital components. The block diagram for this type of control system is given in Fig. 2 below.

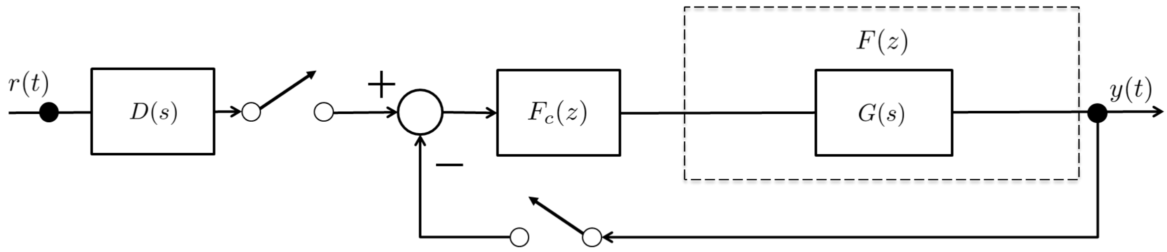

Figure 2: Solar Array Control System

(c) Assuming $F_c(z)$ is derived from the continuous time controller $G_c(s)$ given by (9), find the discrete time controller $F_c(z)$ using the Matched-Zero-Pole (MPZ) method.

$$G_c(s) = 5\,\frac{s+2}{s+8} \tag{9}$$

(d) Why do you think $D(s)$ was included in the control design? What additional components could you include in the control design to improve the control system performance?

QUESTION 5

A sustainable aviation fuels (SAF) company uses digital control hardware technology to help manage production of ethanol from biomass. They have been using an set of difference equations given by (10) to represent the chemical reactions inside the tanks

$$\begin{aligned} \epsilon(kT+T) &= \epsilon(kT) - 2(1-e^{-aT})\theta(kT) && \text{(10a)} \\ \theta(kT+T) &= e^{-aT}\theta(kT) - a^{-1}(1-e^{-aT})u(kT) && \text{(10b)} \\ y(kT+T) &= -12a\theta(kT) && \text{(10c)} \end{aligned}$$

where θ is the temperature, ϵ is the ethanol concentration, u is the input biomass rate (kgs^{-1}) and y is the measured output. The company's ethanol purity has degraded and believe their digital control systems are responsible. They have asked you to help.

(a) Derive a suitable discrete time state space model for the system in (10) assuming a sampling time $T = 2$ seconds and parameter $a = 0.1$.

(b) Provide a brief analysis of the model that would help the company design digital control systems to regulate tank temperature or ethanol concentration.

(c) The company asks you design a digital control system to regulate ethanol concentration. The performance specifications include a natural frequency $w_n = 0.83$ and percentage overshoot of approximately 1.5% when the digital hardware runs at a sampling time $T = 2$ seconds. Design a state feedback controller to meet these specifications.

(d) The company implements your controller, but the performance specifications are not achieved. You request more information and they state the parameter a may be in error of up to 30%.

Diagnose the problem then suggest potential solutions for the company, justifying your response with evidence.